1. (习题十一 15)

取 f 的多项式逼近 $p_k \rightrightarrows f$, 设 $p_k(x) = \sum_{i=0}^{m_k} a_{k,i}x^i$, 则

$$\int_0^1 f(x)p_k(x)\mathrm{d}x = \sum_{i=0}^{m_k} a_{k,i}\int_0^1 f(x)x^i\mathrm{d}x = 0.$$

因此

$$\lim_{k\to\infty}\int_0^1 f(x)p_k(x) = \int_0^1 f(x)\left(\lim_{k\to\infty} p_k(x)\right)\mathrm{d}x = \int_0^1 f(x)^2\mathrm{d}x = 0.$$

所以 $f \equiv 0$.

2. (习题十一 18)

取 f 的多项式逼近 $p_n \rightrightarrows f$. 对任意 $\varepsilon > 0$, 存在 N, 使得 $\|f - p_N\| \le \frac{\varepsilon}{3}$; 同时存在 $\delta > 0$ 使得 $|x-y| < \delta$ 时 $|p_N(x) - p_N(y)| < \frac{\varepsilon}{3}$. 则对任意 $x, y \in (a,b)$, 当 $|x-y| < \delta$:

$$|f(x) - f(y)| \le |f(x) - p_N(x)| + |p_N(x) - p_N(y)| + |p_N(y) - f(y)| < \frac{\varepsilon}{3}\cdot 3 = \varepsilon.$$

所以 f 在 (a,b) 一致连续.

3. (补充题)

> 若一个次数一致有界的多项式列一致收敛到 $f \in C[a,b]$, 证明 f 为多项式.

设多项式列 $p_n \rightrightarrows f$, $\sup \deg p_n = m$, 设 $p_n(x) = \sum_{i=0}^m a_{ni}x^i$, 则有

$$\lim_{n\to\infty} p_n(x) = \lim_{n\to\infty}\sum_{i=0}^m a_{ni}x^i = \sum_{i=0}^m\left(\lim_{n\to\infty} a_{ni}\right)x^i = f(x).$$

只需证明第三式中 $\lim_{n\to\infty} a_{ni} =: a_i$ 总是存在. 在 $[a,b]$ 上任意取定互异的 $x_0, \cdots, x_n$, 设方阵 $V \in \mathbf{M}_{(n+1)\times(n+1)}(\mathbb{R})$ 满足 $V_{ij} = x_i^j$, 则

$$V\begin{pmatrix} a_{n0} \\ \vdots \\ a_{nm}\end{pmatrix} = \begin{pmatrix} p_n(x_0) \\ \vdots \\ p_n(x_n)\end{pmatrix} \Rightarrow \begin{pmatrix} a_{n0} \\ \vdots \\ a_{nm}\end{pmatrix} = V^{-1}\begin{pmatrix} p_n(x_0) \\ \vdots \\ p_n(x_n)\end{pmatrix}.$$

左右对 n 取极限 (或对左侧逐分量取极限以验证):

$$\begin{pmatrix} a_0 \\ \vdots \\ a_n\end{pmatrix} = V^{-1}\begin{pmatrix} f(x_0) \\ \vdots \\ f(x_n)\end{pmatrix},$$

总是存在.

4. (习题十二 1)

(1) f 为偶函数, $b_k \equiv 0$.

$$
\begin{aligned}
a_k &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos kx\mathrm{d}x \\
&= \frac{2}{\pi}\int_{0}^{\pi} x\cos kx\mathrm{d}x \\
&= -\frac{2}{k\pi}\int_{0}^{\pi} \sin kx\mathrm{d}x \\
&= -\frac{2}{k^2\pi}(\cos kx)|_0^{\pi}.
\end{aligned}
$$

因而 $a_0=\pi^2$, $a_{2k+1}=-\frac{4}{k^2\pi}$, $a_{2k}=0$,

$$
f(x)\sim\frac{\pi}{2}-\frac{4}{\pi}\sum_{n=0}^{\infty}\frac{\cos(2n+1)x}{(2n+1)^2}.
$$

(3) f 为奇函数, $a_k=0$.

$$
\begin{aligned}
b_k &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin kx\mathrm{d}x \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos x\sin kx\mathrm{d}x \\
&= -\frac{2}{k\pi}\int_{0}^{\pi} x\cos x\mathrm{d}\cos kx \\
&= -\frac{2}{k\pi}\left(x\cos x\cos kx|_0^{\pi}-\int_0^{\pi}\cos kx\cos x\mathrm{d}x+\int_0^{\pi}x\sin x\cos kx\mathrm{d}x\right).
\end{aligned}
$$

分部积分可知, $k\geq 2$ 时 $\int_0^{\pi}\cos kx\cos x\mathrm{d}x=\int_0^{\pi}\sin kx\sin x\mathrm{d}x=0$. 同理,

$$
\int_0^{\pi}x\sin x\cos kx\mathrm{d}x=-\frac{1}{k}\left(\int_0^{\pi}\sin kx\sin x\mathrm{d}x+\int_0^{\pi}x\cos x\sin kx\mathrm{d}x\right).
$$

所以 $k\geq 2$ 时

$$
b_k=-\frac{2}{k\pi}\left(-\pi\cos k\pi-\frac{\pi}{2k}b_k\right)=\frac{2k(-1)^k}{k^2-1}.
$$

而

$$
b_1=\frac{1}{\pi}\int_0^{\pi}x\sin 2x\mathrm{d}x=\frac{1}{\pi}\left(-\frac{\pi}{2}+\int_0^{\pi}\cos 2x\mathrm{d}x\right)=-\frac{1}{2}.
$$

总之

$$
f(x)\sim-\frac{1}{2}\sin x+\sum_{k=2}^{\infty}\frac{2k(-1)^k}{k^2-1}\sin kx.
$$

(5) $a_k=0$, $b_k=\frac{2}{\pi}\int_0^{\pi}\frac{\pi}{4}\sin kx\mathrm{d}x=\frac{1}{2k}(1-\cos k\pi)$, 则

$$
f(x)\sim\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{2n-1}.
$$

(7)

$$
\begin{aligned}
f(x) &= \frac{1-r^2}{1-2r\cos x+r^2} \\
&= \frac{1-r^2}{1-r(\mathrm{e}^{\mathrm{i}x}+\mathrm{e}^{-\mathrm{i}x})+r^2} \\
&= \frac{(1-r^2)}{(1-r\mathrm{e}^{\mathrm{i}x})(1-r\mathrm{e}^{-\mathrm{i}x})} \\
&= (1-r^2)\left(\sum_{n=0}^{\infty}(r\mathrm{e}^{\mathrm{i}x})^n\right)\left(\sum_{n=0}^{\infty}(r\mathrm{e}^{-\mathrm{i}x})^n\right) \\
&= (1-r^2)\sum_{k=-\infty}^{\infty}\frac{r^{|k|}}{1-r^2}\mathrm{e}^{\mathrm{i}kx} \\
&= 1+\sum_{k=1}^{\infty}r^k(\mathrm{e}^{\mathrm{i}kx}+\mathrm{e}^{-\mathrm{i}kx}) \\
&= 1+2\sum_{k=1}^{\infty}r^k\cos kx.
\end{aligned}
$$

因此

$$
f(x) \sim 1+\sum_{k=1}^{\infty}2r^k\cos kx.
$$

5. (习题十二 2)

(1) 正弦级数, 延拓为 $f(x)=\sin x$, $f(x)\sim\sin x$.

余弦级数, 延拓为 $f(x)=|\sin x|$, $a_k=\frac{2}{\pi}\int_0^{\pi}\sin x\cos kx\mathrm{d}x$, 则

$$
a_k=\begin{cases}0, & 2\nmid k;\\ \frac{4}{(1-k^2)\pi}, & 2\mid k.\end{cases}
$$

$$
f(x)\sim\frac{2}{\pi}-\sum_{n=1}^{\infty}\frac{4\cos 2nx}{(4n^2-1)\pi}.
$$

(3) 正弦级数,

$$
\begin{aligned}
b_k &= \frac{2}{\pi}\int_0^{\pi}x(\pi-x)\sin kx\mathrm{d}x \\
&= 2\int_0^{\pi}x\sin kx\mathrm{d}x-\frac{2}{\pi}\int_0^{\pi}x^2\sin kx\mathrm{d}x \\
&= -\frac{2}{k}\left(x\cos kx\big|_0^{\pi}-\int_0^{\pi}\cos kx\mathrm{d}x\right)+\frac{2}{k\pi}\left(x^2\cos kx\big|_0^{\pi}-2\int_0^{\pi}x\cos kx\mathrm{d}x\right) \\
&= -\frac{2}{k}\pi\cos k\pi+\frac{2}{k\pi}\left(\pi^2\cos k\pi-\frac{2}{k^2}(\cos k\pi-1)\right) \\
&= -\frac{4(\cos k\pi-1)}{k^3\pi}.
\end{aligned}
$$

$$
f(x)\sim\sum_{n=1}^{\infty}\frac{8}{(2n-1)^3\pi}\sin(2n-1)x.
$$

余弦级数, $a_0=\frac{\pi^2}{3}$,

$$
\begin{aligned}
a_k &= \frac{2}{\pi}\int_0^{\pi} x(\pi - x)\cos kx dx \\
&= 2\int_0^{\pi} x\cos kx\mathrm{d}x - \frac{2}{\pi}\int_0^{\pi} x^2\cos kx\mathrm{d}x \\
&= \frac{2}{k}\left(x\sin kx|_0^{\pi} - \int_0^{\pi}\sin kx\mathrm{d}x\right) - \frac{2}{k\pi}\left(x^2\sin kx\Big|_0^{\pi} - 2\int_0^{\pi} x\sin kx\mathrm{d}x\right) \\
&= -\frac{2(1+\cos k\pi)}{k^2}.
\end{aligned}
$$

$$
f(x) \sim \frac{\pi^2}{6} - \sum_{n=1}^{\infty}\frac{\cos 2nx}{n^2}.
$$

6. (习题十二 5)

不妨设 $f(x) = g(x) - h(x)$, 其中 g, h 均单增. 由于 f 在闭区间 $[0, 2\pi]$ 有定义, g, h 均有界, 均 Riemann 可积. 因此

$$
\begin{aligned}
\pi a_k &= \int_0^{2\pi}(g(x) - h(x))\cos kx\mathrm{d}x \\
&= \int_0^{2\pi} g(x)\cos kx\mathrm{d}x - \int_0^{2\pi} h(x)\cos kx\mathrm{d}x.
\end{aligned}
$$

其中

$$
\int_0^{2\pi} g(x)\cos kx\mathrm{d}x = \frac{1}{k}\left(g(x)\sin kx|_0^{2\pi} - \int_0^{2\pi}\sin kx\mathrm{d}g\right) = -\frac{1}{k}\int_0^{2\pi}\sin kx\mathrm{d}g.
$$

所以

$$
|\pi k a_k| = \left|\int_0^{2\pi}\sin kx\mathrm{d}g\right| \le \int_0^{2\pi}\mathrm{d}g = O(1).
$$

a_k 的另一项和 b_k 均同理.